



Homework 8

Deadline: 9 December 2025, 23:30.

All solutions must be in a single PDF file and uploaded to the LMS portal.

1. Consider a vanilla call option on an underlying asset S with expiration date T and strike K under Black-Scholes-Merton model.
 - (a) (0.5 point) At what price of the underlying asset is the gamma of this option maximized at time t ? Justify your answer.
 - (b) (0.5 point) Find the value of t at which the gamma reaches its maximum, given that the price of the underlying asset is equal to the option's strike price.
2. (1 point) Let $x \in \mathbb{R}^n$, and consider a function $f(x, z) \in \mathbb{R}$ where $z \in \mathbb{R}^m$. Let $g(x) \in \mathbb{R}^m$ be a vector function of a vector argument and define the composite function $h(x) = f(x, g(x))$, so that $h(x) \in \mathbb{R}$. Using the notation from the lecture, prove that the following holds:

$$\frac{\partial^2 h}{\partial x^2} \neq \frac{\partial^2 f}{\partial x^2}.$$

What additional terms must be included on the right-hand side to obtain equality? Justify your answer.